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#include <Adafruit_Sensor.h>

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// Experiment 7: Stepper Motor Control via Microcontroller and LCD
// showing Speed and Direction

#include <Wire.h>
#include <Stepper.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
const int stepsPerRevolution = 2048;
Stepper myStepper(stepsPerRevolution, 8, 10, 9, 11);

void setup() {
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Stepper Test");
  delay(1000);
  lcd.clear();

  Serial.println("Stepper Motor Speed Test with I2C LCD");
}
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}

void loop() {
  for (int spd = 5; spd <= 10; spd += 5) {
    myStepper.setSpeed(spd);

    Serial.print("Rotating CCW at ");
    Serial.print(spd);
    Serial.println(" RPM");

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Direction: CCW");
    lcd.setCursor(0, 1);
    lcd.print("Speed: ");
    lcd.print(spd);
    lcd.print(" RPM");

    myStepper.step(stepsPerRevolution / 2); // Half revolution CW
    delay(500);
  }

  for (int spd = 10; spd >= 5; spd -= 5) {
    myStepper.setSpeed(spd);

    Serial.print("Rotating CW at ");

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Serial.print(spd);  
Serial.println(" RPM");  
  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("Direction: CW");  
lcd.setCursor(0, 1);  
lcd.print("Speed: ");  
lcd.print(spd);  
lcd.print(" RPM");  
  
myStepper.step(-stepsPerRevolution / 2);  
delay(500);  
}  
}
```